Spatial Light Modulation
Experiment 04

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1 Aim

To measure the linear retardance and optical rotation of a SLM device for various grey scale values.

2 Apparatus

The required apparatus for performing the experiment are:

1. Laser source
2. SLM device
3. Polarizers
4. Glass Slab
5. Neutral-density filter
6. Current detector

3 Theory

3.1 Polarization of Light

Light is a transverse electromagnetic wave. For a wave travelling in in the $\hat{z}$ direction we have,

$$E_x = E_{0x}e^{i(\omega t - kz + \delta_x)}, \quad E_y = E_{0y}e^{i(\omega t - kz + \delta_y)} \tag{1}$$

Depending on the amplitudes and phase difference between the components of the electric field, we define different polarizations viz. linear, circular, or elliptical.

3.2 Stokes Parameters and Mueller Matrices

The complete description of the polarization state is given by the Stokes vector, based on the intensity parameters:

$$S = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} I_H + I_V \\ I_H - I_V \\ I_P - I_M \\ I_R - I_L \end{bmatrix} \tag{2}$$

where S_0 is the total intensity of the beam, S_1 represents the difference of intensities in the horizontal or vertical polarization $(I_H - I_V)$, S_2 denotes the difference in intensities between $+45°$ or $-45°$ polarization states $((I_P - I_M))$ while S_3 represents the difference in intensities of the circular polarization states $(I_R - I_L)$. The interaction of polarized light with an optical element can be described by the Mueller matrix M:

$$S_{\text{out}} = \text{M}S_{\text{in}} \qquad M \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \tag{3}$$

where S_{in} and S_{out} are the input and output Stokes vectors.

3.2.1 Mueller Matrix of a Linear Retarder

An important property of an optical medium is *retardance*, which arises due to the difference in the refractive index of two orthogonal polarization states and results in a phase difference. For a linear retarder with retardance δ and fast axis at angle θ:

$$M_{\text{lin}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\theta + \cos\delta \sin^2 2\theta & (1-\cos\delta)\sin 2\theta \cos 2\theta & -\sin\delta \sin 2\theta \\ 0 & (1-\cos\delta)\sin 2\theta \cos 2\theta & \sin^2 2\theta + \cos\delta \cos^2 2\theta & \sin\delta \cos 2\theta \\ 0 & \sin\delta \sin 2\theta & -\sin\delta \cos 2\theta & \cos\delta \end{bmatrix} \tag{4}$$

3.2.2 Mueller Matrix of a Circular Retarder (Optical Rotator)

For optical rotation by an angle ψ, the Mueller matrix becomes:

$$M_{\text{circ}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\psi & \sin 2\psi & 0 \\ 0 & -\sin 2\psi & \cos 2\psi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

3.2.3 Mueller Matrix of SLM

Spatial Light Modulator is a optoelectronic device, containing twisted layers of liquid crystals and is used to modulate the magnitude or phase of a beam of light. Electric field can be applied to tilt these crystals to produce the intended effect. SLM behaves as a combination of a linear retarder and an optical rotator:

$$M_{\text{SLM}} = M_{\text{circ}} \cdot M_{\text{lin}} \tag{6}$$

Thus, the SLM introduces both retardance (δ) and optical rotation (ψ), which are determined experimentally using Stokes measurements. For horizontal incident light, the output Stokes vector is,

$$S_1 = \begin{bmatrix} 1 \\ \cos 2\theta \cos(2\theta - 2\psi) + \sin 2\theta \cos\delta \sin(2\theta - 2\psi) \\ \sin 2\theta \cos(2\theta - 2\psi) - \cos 2\theta \cos\delta \sin(2\theta - 2\psi) \\ \sin(2\theta - 2\psi) \end{bmatrix} \tag{7}$$

For the +45° polarization, the output Stokes vector is given by,

$$S_2 = \begin{bmatrix} 1 \\ \cos 2\theta \sin(2\theta - 2\psi) - \sin 2\theta \cos\delta \cos(2\theta - 2\psi) \\ \sin 2\theta \sin(2\theta - 2\psi) + \cos 2\theta \cos\delta \cos(2\theta - 2\psi) \\ -\cos(2\theta - 2\psi)\sin\delta \end{bmatrix} \tag{8}$$

From this we calculate the linear retardance δ and the rotation ψ of an SLM by,

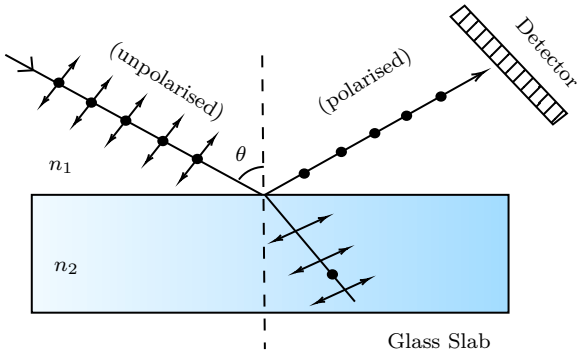
$$\begin{aligned} [S_1]_2 + [S_2]_3 &= (1+\cos\delta)\cos 2\psi = X \\ [S_1]_3 - [S_2]_2 &= (1+\cos\delta)\sin 2\psi = Y \end{aligned} \tag{9}$$

where $[S_i]_j$ is the jth component of S_i vector. Simplifying this gives us,

$$\delta = \cos^{-1}\left[\sqrt{X^2+Y^2} - 1\right] \qquad \psi = \tan^{-1}\left(\frac{Y}{X}\right) \tag{10}$$

4 Brewster's Angle

To determine the polarization axis of the given polarizer, we need to use the concept of Brewster's angle, which is the angle of incidence at which light of a particular polarization will transmit fully through a dielectric surface. If unpolarized light is incident, it becomes polarized after reflection from the surface.



If light is going from a medium with refective index n_1 to a medium of refractive index n_2 then Brewster's angle is given by,

$$\theta_{\mathrm{B}} = \tan^{-1}\left(\frac{n_1}{n_2}\right) \tag{11}$$

The experiment consists of two parts, first we determine the polarisation axis of the two polarizers using Brewster's angle.

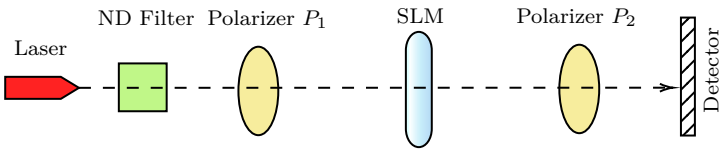


Figure 1: Schematic of the Experimental Setup

In the second part, we find the intensities for different grayscale values of the SLM and try to reconstitute the Stokes vector.

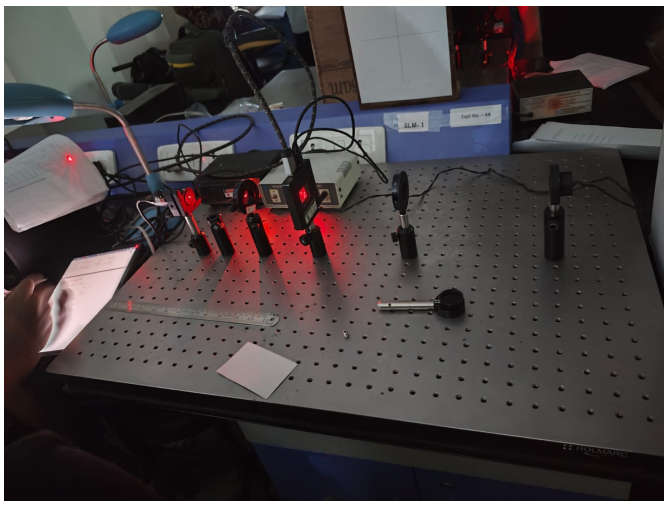


Figure 2: Actual Experimental Setup

5 Data

5.1 Calibration of Polarizer Axis

The reference angular positions corresponding to horizontal, vertical and the $\pm 45°$ polarization states by determination of *Brewster's* angle. Allowing the laser to fall on a air-glass interface, the angle of incidence was gradually varied while monitoring the reflected intensity. The angle corresponding to minimum reflected intensity was identified as *Brewster's* angle, when the reflected beam becomes purely s-polarized, allowing identification of the horizontal axis of P_1. Then we placed the second polarizer in front of the first, to determine the vertical angular position of the second. The data is tabulated below.

	P_1	P_2
H	$284°$	$222° - 90° = 132°$
V	$284 + 90° = 374° \equiv 14°$	$222°$
P	$284 + 45° = 329°$	$132° + 45° = 177°$
M	$284 - 45° = 239°$	$132° - 45° = 87°$

Table 1: Polarization angle for P_1 and P_2

5.2 Measuring Polarization of SLM

After calibration, we put the SLM in between the polarizers. Then we varied the grey scale and for each horizontal and vertical position of P_1, we aligned P_2 into the four polarization states and noted the intensity of the transmitted light. The data is tabulated below.

Grey Value	Input (P_1)	I_H	I_V	I_P	I_M
0	H	3.9	5.1	3.6	1.6
	P	16.8	40.8	2.4	51.4
64	H	0.8	2.4	2.5	1.3
	P	16.6	19.1	17.0	32.2
128	H	0.4	4.0	2.7	1.5
	P	12.3	39.2	2.6	50.0
255	H	0.5	3.7	2.7	1.4
	P	15.3	39.6	15.3	50.5

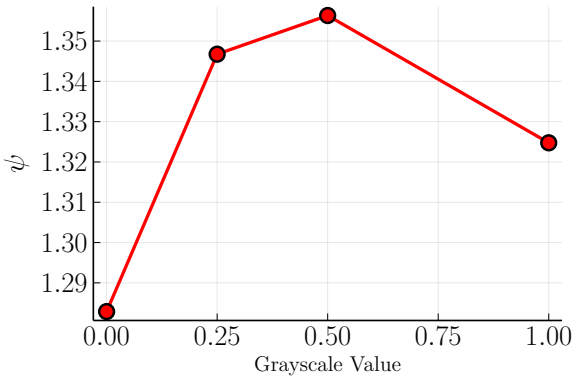
Table 2: Photodetector current for different grey values and polarization states.

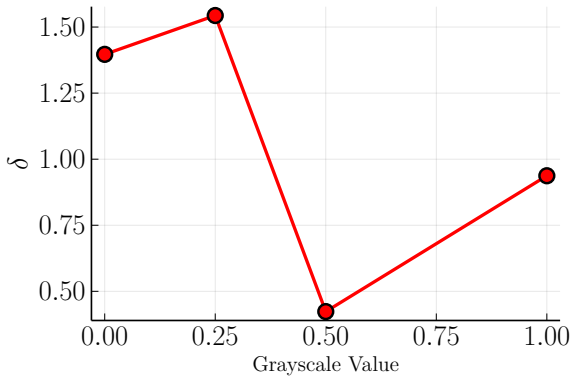
We calculate the Stokes vector from the data above, according to the given formulas.

Grayscale	$\mathbf{P_1}$	**Normalized Stokes Vector S**
0	Horizontal (H)	$\begin{bmatrix} 1 \\ -0.133 \\ 0.222 \\ - \end{bmatrix}$
0	+45°State (P)	$\begin{bmatrix} 1 \\ -0.416 \\ -0.850 \\ - \end{bmatrix}$
64	Horizontal (H)	$\begin{bmatrix} 1 \\ -0.5 \\ 0.375 \\ - \end{bmatrix}$
64	+45°State (P)	$\begin{bmatrix} 1 \\ -0.07 \\ -0.426 \\ - \end{bmatrix}$
128	Horizontal (H)	$\begin{bmatrix} 1 \\ -0.818 \\ 0.273 \\ - \end{bmatrix}$
128	+45°State (P)	$\begin{bmatrix} 1 \\ -0.522 \\ -0.920 \\ - \end{bmatrix}$
255	Horizontal (H)	$\begin{bmatrix} 1 \\ -0.761 \\ 0.309 \\ - \end{bmatrix}$
255	+45°State (P)	$\begin{bmatrix} 1 \\ -0.443 \\ -0.641 \\ - \end{bmatrix}$

Table 3: Normalized Stokes Vectors for Different Grayscale values

We then calculated X and Y, from which we obtained δ and ψ, using the formulas mentioned before and then the extracted linear retardance and optical rotation were plotted as functions of greyscale values.





6 Error Analysis

6.1 Random Error

We analyze the errors introduced in the results. The Stokes' vectors were defined in Eq. 2. Using this, the uncertainty in the second component of the Stokes vector becomes,

$$\Delta[S]_2 = \left|\frac{2I_V}{(I_H+I_V)^2}\right|\Delta I + \left|\frac{2I_H}{(I_H+I_V)^2}\right|\Delta I = \frac{2\Delta I}{(I_H+I_V)} \tag{12}$$

Simularly, for the third component we have the uncertainty

$$\Delta[S]_3 == \frac{2\Delta I}{(I_H+I_V)} + \frac{2|I_P - I_M|\Delta I}{(I_H+I_V)^2} \tag{13}$$

Since X and Y are linearly related to these components, we get the uncertainty in X and Y just by adding these uncertainities. From that, we obtain the uncertainty in ψ and δ.

$$\psi = \frac{1}{2}\tan^{-1}\left(\frac{Y}{X}\right) \implies \frac{\partial\psi}{\partial X} = -\frac{Y}{2(X^2+Y^2)} \quad \frac{\partial\psi}{\partial Y} = \frac{X}{2(X^2+Y^2)} \implies \Delta\psi = -\frac{|Y|\Delta X}{2(X^2+Y^2)} + \frac{|X|\Delta Y}{2(X^2+Y^2)}$$

Similarly, if we do this for the linear retardance we get,

$$\delta = \cos^{-1}\left(\sqrt{X^2+Y^2}-1\right) \implies \Delta\delta = \frac{|X|\Delta X + |Y|\Delta Y}{\sqrt{X^2+Y^2}\sqrt{1-(\sqrt{X^2+Y^2}-1)^2}}$$

The least count of the photodetector was 0.1 μA which implies $\Delta I = 0.1\ \mu A$. Using these formula above, we get the maximum possible errors for each greyscale value as,

Grayscale	$\Delta\psi$ (rad.)	$\Delta\delta$ (rad.)
0	0.016	0.036
64	0.054	0.098
128	0.02	0.163
255	0.026	0.09

6.2 Sources of Error

- The laser was not stable at all times which could have led to fluctuations in the measured internsity.
- The polarizers were not perfect, hence there was always some residual component from the unintended polarization states which could have introduced errors.
- The SLM crystals might have some spatial non-uniformity which could have caused error in measurement.
- The variation in external temperature could have affected the SLM crystals, leading to some error.

7 Discussion and Conclusion

The variation of linear retardance (δ) and optical rotation (ψ) with grayscale demonstrates that the SLM effectively modulates the polarization state of light. The observed dependence is clearly non-linear, which is expected due to the nonlinear electro-optic response of liquid crystal molecules to the applied voltage.

The retardance shows a smooth variation with grayscale, indicating controlled phase modulation. In contrast, the optical rotation exhibits a non-monotonic behavior, reflecting the complex interplay between birefringence and molecular reorientation within the SLM.

Uncertainty analysis shows that the error in δ is significantly larger than that in ψ. This arises from the higher sensitivity of the inverse cosine function used in calculating retardance, making it more susceptible to small fluctuations in measured intensities.

Deviations from ideal behavior can be attributed to experimental limitations such as polarizer misalignment, finite extinction ratio, laser intensity fluctuations, and non-uniformity of SLM pixels. Despite these factors, the overall trends remain consistent and physically reasonable.

In this experiment, we studied the polarization properties of a SLM using Stokes polarimetry. The linear retardance (δ) and optical rotation (ψ) were extracted for different grayscale values from measured intensity data.

The measured values indicate that the SLM introduces significant phase retardation and polarization rotation, both of which vary with the applied grayscale. The estimated uncertainties were $\Delta\psi \approx 0.016$ to $\Delta\psi \approx 0.054$ rad and $\Delta\delta \approx 0.036$ to $\delta \approx 0.163$ rad within the values of the grayscale measured. The larger uncertainty in δ is consistent with the higher sensitivity of the inverse cosine function used in its calculation.

Overall, the results confirm that the SLM acts as a voltage-controlled birefringent element capable of modulating both phase and polarization of light. The observed trends and magnitudes are consistent with the expected nonlinear electro-optic response of liquid crystal systems.